

PART I - FOUNDATION PLC PROJECTS

Project 5 - Conveyor System

Version 1 - Foundation PLC Project

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite 8.30

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1. Project Overview

The Conveyor System was developed to combine motor control, product counting, and batch-complete indication in one PLC application. The system allows an operator to start and stop a conveyor, count boxes only while the conveyor is running, and indicate when a batch of 10 boxes has been completed.

Start Push Button I:1/0 latches Conveyor Motor O:2/0 ON, while Stop Push Button I:1/1 unlatches the same output. Box Sensor I:1/2 is used with the conveyor-running condition to drive Counter C5:0. When the accumulated count reaches the preset of 10, Batch Complete Lamp O:2/1 energizes.

Reset Push Button I:1/3 executes RES C5:0, clearing the accumulated count and Done status so a new batch can begin.

This project expanded my PLC foundation by combining maintained motor control with event-based counting and a simple production-batch indication.

2. Engineering Objective

The objective of this project was to develop and verify a PLC program that controlled a conveyor motor, counted boxes only while the conveyor was operating, indicated when 10 boxes had been counted, and allowed the operator to reset the batch counter.

- Start and maintain conveyor operation from an operator command.
- Stop the conveyor from a separate operator command.
- Count valid box-detection events only when the conveyor is running.
- Indicate completion when the counter reaches a preset of 10.
- Reset the counter and batch-complete indication for the next cycle.

3. Functional Requirements

- The PLC shall monitor Start Push Button I:1/0.
- When I:1/0 becomes true, Conveyor Motor O:2/0 shall be latched ON.
- The PLC shall monitor Stop Push Button I:1/1.
- When I:1/1 becomes true, Conveyor Motor O:2/0 shall be unlatched OFF.
- Counter C5:0 shall count box-detection events only when Conveyor Motor O:2/0 is ON.
- Box Sensor I:1/2 shall provide the counting event.
- Counter C5:0 shall use a preset of 10.
- Batch Complete Lamp O:2/1 shall remain OFF while C5:0/DN is false.
- Batch Complete Lamp O:2/1 shall energize when C5:0/DN becomes true.
- Reset Push Button I:1/3 shall execute RES C5:0.
- Resetting C5:0 shall return the accumulator to zero, clear the Done bit, and turn the Batch Complete Lamp OFF.
- The conveyor shall not stop automatically when the counter reaches 10; it shall remain latched ON until the Stop Push Button is activated.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes the conveyor and counting logic
Input Module	1762-IQ16	Provides discrete operator and sensor inputs
Output Module	1762-OB16	Provides conveyor and lamp outputs
Start Push Button	I:1/0	Starts/latches the conveyor
Stop Push Button	I:1/1	Stops/unlatches the conveyor
Box Sensor	I:1/2	Provides box-detection events
Reset Push Button	I:1/3	Resets the batch counter
Conveyor Motor	O:2/0	Represents conveyor drive command
Batch Complete Lamp	O:2/1	Indicates that the preset count has been reached
Box Counter	C5:0	Maintains batch count
Programming Software	RSLogix Micro Starter Lite 8.30	Ladder logic development and monitoring

This Version 1 project models the conveyor control logic. It does not include a VFD, motor overload feedback, emergency-stop circuit, or higher-level production system.

5. I/O Assignment

Physical I/O

Device	PLC Address	Type	Description
Start Push Button	I:1/0	Digital Input	Starts/latches conveyor
Stop Push Button	I:1/1	Digital Input	Stops/unlatches conveyor
Box Sensor	I:1/2	Digital Input	Detects each box-counting event
Reset Push Button	I:1/3	Digital Input	Resets batch counter
Conveyor Motor	O:2/0	Digital Output	Conveyor motor command
Batch Complete Lamp	O:2/1	Digital Output	Batch-complete indication

Internal PLC Data

Function	Address	Type	Purpose
Box Counter	C5:0	CTU Counter	Stores accumulated box count
Counter Preset	C5:0.PRE	Counter Data	Target quantity of 10
Counter Accumulator	C5:0.ACC	Counter Data	Current box count
Counter Done	C5:0/DN	Status Bit	Becomes true when count reaches preset

6. Control Strategy

I used a five-function control strategy: conveyor start, conveyor stop, box counting, batch-complete indication, and counter reset.

Start Push Button I:1/0 executes OTL O:2/0 so the conveyor remains commanded ON after the Start button is released. Stop Push Button I:1/1 executes OTU O:2/0 to remove the latched conveyor command.

The counting rung requires both Conveyor Motor O:2/0 and Box Sensor I:1/2 to be true. This ensures that a sensor transition cannot increment the counter while the conveyor is stopped.

Counter C5:0 uses a preset of 10. When the accumulated count reaches the preset, C5:0/DN becomes true and energizes Batch Complete Lamp O:2/1.

Reset Push Button I:1/3 executes RES C5:0 to clear the current batch count. Resetting the counter does not stop the conveyor.

7. Engineering Design Decisions

Design Decision	Reason
Used OTL O:2/0 for conveyor start	Allowed the conveyor to remain ON after a momentary Start command
Used OTU O:2/0 for conveyor stop	Provided a separate command to remove the latched conveyor state
Placed Conveyor Motor status ahead of the Box Sensor	Prevented counting while the conveyor was stopped
Used CTU C5:0	The application required event-based product counting
Set counter preset to 10	Defined the required batch size
Used C5:0/DN for Batch Complete Lamp	The Done bit directly represents reaching the target quantity
Used RES C5:0 for batch reset	Allowed the next batch to begin from zero
Did not automatically stop the conveyor at count 10	Version 1 required indication only; automatic batch stop is reserved for Version 2

A limitation of this design is that the physical conveyor output O:2/0 is directly latched and unlatched. Version 2 can improve this by using an internal Run Request and controlling the physical output through permissives, interlocks, and faults.

8. Ladder Logic Description

The Version 1 program contains five functional rungs. Each rung is presented below using the approved single-line ladder format.

8.1 Rung 0000 - Start / Latch Conveyor

```
|----[ XIC I:1/0 ]----- ( OTL O:2/0 )----|
      Start_PB              Conveyor_Motor
```

When Start Push Button I:1/0 becomes true, OTL O:2/0 latches the Conveyor Motor ON. The conveyor remains commanded ON after the Start button is released.

8.2 Rung 0001 - Stop / Unlatch Conveyor

```
|----[ XIC I:1/1 ]----- ( OTU O:2/0 )----|
      Stop_PB              Conveyor_Motor
```

When Stop Push Button I:1/1 becomes true, OTU O:2/0 unlatches the Conveyor Motor and stops the conveyor.

8.3 Rung 0002 - Count Boxes While Conveyor Runs

```
|----[ XIC O:2/0 ]----[ XIC I:1/2 ]-----[ CTU C5:0 ]----|
      Conveyor_Motor   Box_Sensor           Box_Counter   PRE: 10
```

The counter rung is enabled only while Conveyor Motor O:2/0 is ON. A valid false-to-true transition of the counting rung increments C5:0 by one. If the conveyor is OFF, sensor activity cannot increment the counter.

8.4 Rung 0003 - Batch Complete Indication

```
|-----[ XIC C5:0/DN ]------( OTE O:2/1 )-----|
      Box_Counter_Done              Batch_Complete_Lamp
```

When C5:0.ACC reaches the preset of 10, C5:0/DN becomes true and energizes Batch Complete Lamp O:2/1. The conveyor remains latched ON until the Stop Push Button is activated.

8.5 Rung 0004 - Reset Batch Counter

```
|-----[ XIC I:1/3 ]-----[ RES C5:0 ]-----|
      Reset_PB              Box_Counter
```

When Reset Push Button I:1/3 becomes true, RES C5:0 returns the accumulated count to zero and clears the Done status. This turns the Batch Complete Lamp OFF and prepares the counter for a new batch.

8.6 Actual RSLogix Implementation

The figure below shows the corresponding ladder logic as implemented in RSLogix Micro Starter Lite.

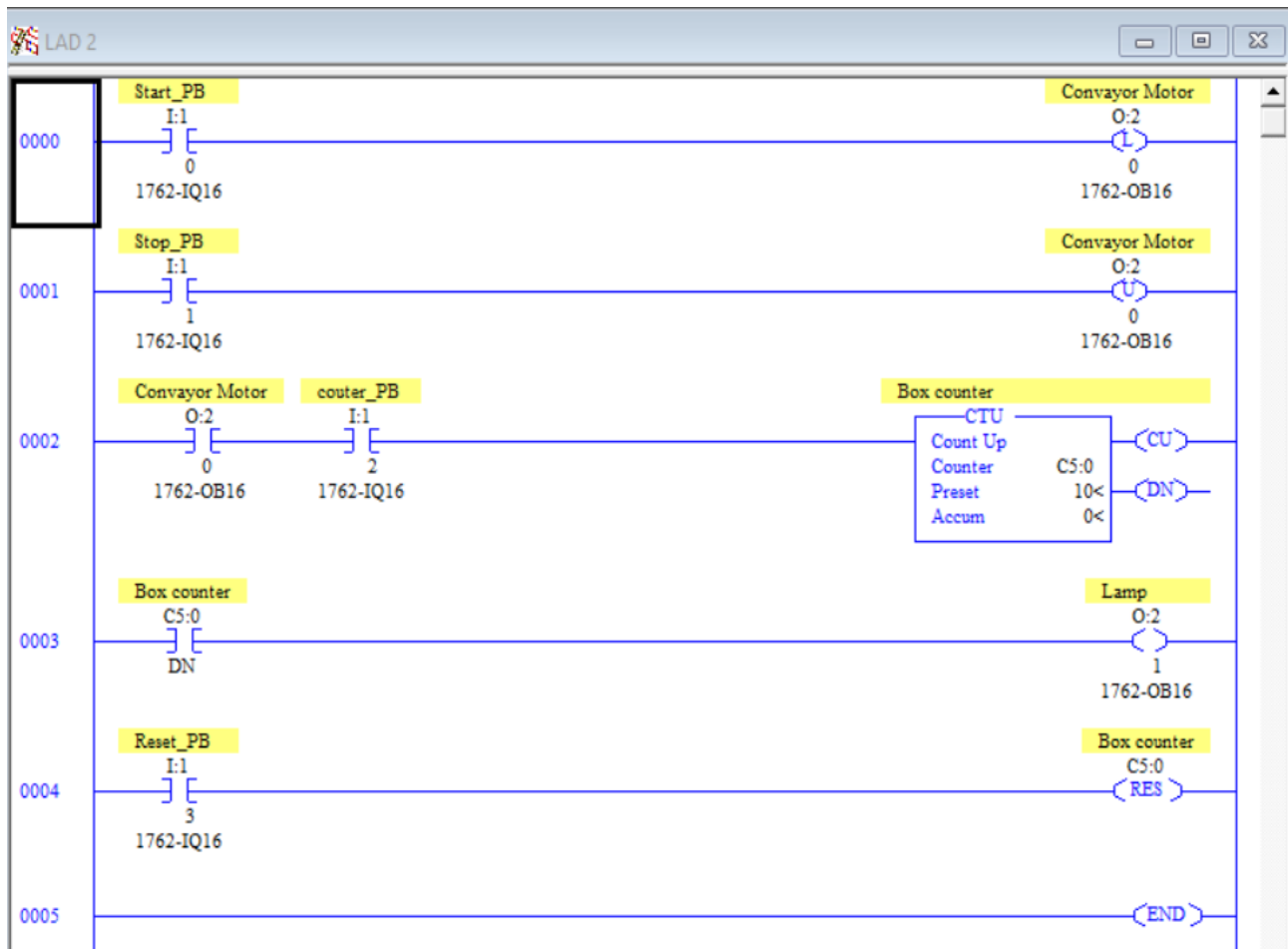


Figure P5.1 - RSLogix Implementation of the Version 1 Conveyor System

9. Sequence of Operation

Step 1 - Initial Condition: Conveyor is OFF, counter is at its current/reset state, and Batch Complete Lamp is OFF when C5:0/DN is false.

Step 2 - Start Command: The operator presses Start Push Button I:1/0.

Step 3 - Conveyor Starts: OTL O:2/0 latches the Conveyor Motor ON.

Step 4 - Box Detection: A box produces a sensor transition at I:1/2 while the conveyor is running.

Step 5 - Counter Increment: C5:0 increments by one.

Step 6 - Continued Counting: Additional valid sensor transitions continue increasing C5:0 while the conveyor remains ON.

Step 7 - Batch Target Reached: When C5:0.ACC reaches 10, C5:0/DN becomes true.

Step 8 - Batch Complete Indication: Batch Complete Lamp O:2/1 turns ON.

Step 9 - Reset Batch Counter: The operator presses Reset Push Button I:1/3; C5:0 resets to zero and the lamp turns OFF.

Step 10 - Conveyor Continues or Stops: The conveyor may continue for the next batch or can be stopped using Stop Push Button I:1/1.

10. Testing & Validation

I tested conveyor start/stop operation, conveyor-dependent counting, batch completion, and counter reset. I also verified that the Box Sensor could not increment the counter while the conveyor was stopped. All six defined Version 1 functional tests passed.

Test ID	Test Action	Expected Result	Observed Result	Status
T-01	Press Start Push Button I:1/0	Conveyor Motor O:2/0 latches ON	Conveyor turned ON and remained latched	PASS
T-02	Press Stop Push Button I:1/1	Conveyor Motor O:2/0 unlatches OFF	Conveyor turned OFF	PASS
T-03	Conveyor ON + valid Box Sensor transition	C5:0 increments by one	Counter incremented correctly	PASS
T-04	Conveyor OFF + Box Sensor transition	C5:0 does not increment	Counter remained unchanged	PASS
T-05	Continue valid counts until C5:0.ACC = 10	C5:0/DN becomes true; Batch Complete Lamp ON	Done bit became true and lamp energized	PASS
T-06	Press Reset Push Button I:1/3	Counter returns to zero; Done clears; lamp OFF	Counter reset and lamp turned OFF	PASS

Validation Summary

Testing confirmed that the conveyor could be started and stopped through the latch/unlatch logic. Box counting occurred only when the conveyor was running, validating the purpose of the O:2/0 permissive in Rung 0002.

When the count reached 10, the Batch Complete Lamp energized as expected. Resetting C5:0 cleared the accumulated count and Done status without stopping the conveyor. All six defined Version 1 functional tests passed.

The testing represents functional PLC program validation within the project development environment, not industrial commissioning or safety validation.

11. Results

The completed Conveyor System successfully performed the required motor control, product counting, batch indication, and reset functions.

The conveyor remained latched ON after the Start command and stopped only when the Stop command was activated. Counter C5:0 incremented only when the conveyor was running and the Box Sensor produced a valid transition.

The Batch Complete Lamp energized when the count reached 10, and the Reset command cleared the count and indication for the next batch. All six functional tests passed.

12. Challenges & Troubleshooting

A useful troubleshooting path for this project is: Start/Stop Inputs -> Conveyor Latched State -> Box Sensor -> Counter Accumulator -> Counter Done Bit -> Batch Complete Lamp.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
Conveyor does not start	Start input or latch rung is not true	Monitor I:1/0 and Rung 0000	Verify Start input and OTL O:2/0
Conveyor does not stop	Stop input or unlatch rung is not true	Monitor I:1/1 and Rung 0001	Verify Stop input and OTU O:2/0
Counter does not increment	Conveyor is OFF or Box Sensor does not transition	Monitor O:2/0, I:1/2, and C5:0.ACC	Verify conveyor status and sensor event
Counter increments unexpectedly	Sensor produces multiple transitions while conveyor runs	Monitor I:1/2 online	Investigate sensor chatter or signal behavior
Batch lamp does not turn ON	Counter has not reached preset or Done bit is incorrect	Monitor C5:0.ACC and C5:0/DN	Verify preset and Rung 0003
Counter does not reset	Reset input or RES instruction is incorrect	Monitor I:1/3 and Rung 0004	Verify Reset input and RES C5:0
Lamp remains ON after Reset	Done bit did not clear or output logic is incorrect	Monitor C5:0/DN and O:2/1	Verify counter reset and lamp rung

One of the main lessons from troubleshooting this project was that product counting depends on both process state and sensor events. A valid sensor transition should not be counted if the conveyor is not running.

13. Lessons Learned

This project helped me combine maintained motor control with event-based production counting in a single PLC program.

I learned how a process-state condition such as Conveyor Motor ON can be used as a permissive for a counter rung.

The project reinforced my understanding of OTL/OTU behavior, CTU transition counting, counter Done status, and RES operation.

I also learned that a batch-complete indication does not automatically mean the process must stop. In this Version 1 design, the lamp indicates completion while conveyor stopping remains a separate operator action.

14. Industrial Relevance

Conveyor-based product counting is common in manufacturing production lines, packaging systems, warehouse automation, food processing, pharmaceutical production, parcel sorting, and assembly operations.

The Version 1 project demonstrates the foundation of a material-handling sequence in which conveyor status is combined with product detection and batch counting.

A real industrial conveyor system would normally include additional features such as motor overload feedback, emergency-stop and safety circuits, jam detection, sensor diagnostics, upstream/downstream interlocks, speed control, batch stopping, operator modes, fault handling, and production data reporting.

For that reason, this project should be viewed as a foundation conveyor-control application rather than a production-ready material-handling system.

15. Transition to Industrial Version (Version 2)

The Version 2 project can retain the basic conveyor, sensor, and batch-counting requirements while improving control structure, diagnostics, and operator interaction.

Version 2 Improvement	Engineering Benefit
Internal Run Request	Separates maintained operating state from the physical conveyor output
Auto/Manual operating modes	Supports normal production and maintenance control
Mode-conflict detection	Identifies invalid simultaneous Auto/Manual commands
Conveyor permissives	Prevents operation unless required conditions are satisfied
Motor overload feedback	Detects drive/motor protection trips
Emergency-stop / safety interface considerations	Keeps standard control logic separate from safety-related shutdown
Sensor diagnostics	Detects stuck or abnormal box-sensor behavior
Jam detection	Identifies product flow problems
Automatic stop at batch completion	Prevents overproduction when the target quantity is reached
Controlled batch reset	Prevents unintended reset during active production
Adjustable batch target	Allows authorized quantity changes without modifying ladder logic
HMI count and status display	Provides real-time operator visibility
Fault/alarm handling	Provides defined responses to abnormal conditions
Production statistics / SCADA integration	Supports higher-level production monitoring

One important structural improvement will be replacing direct OTL/OTU control of O:2/0 with an internal Run Request. The physical conveyor output can then be controlled through operating mode, permissives, interlocks, and fault conditions.

A stronger Version 2 sequence could become: Start Request -> Run Request -> Mode / Permissive / Fault Checks -> Conveyor Command -> Product Counting -> Batch Complete -> Controlled Stop / Reset -> HMI & Diagnostics.

16. Project Summary

The Conveyor System combined latched motor control, conveyor-dependent product counting, batch-complete indication, and counter reset in one PLC application.

Start Push Button I:1/0 latched Conveyor Motor O:2/0 ON, while Stop Push Button I:1/1 unlatched it OFF. Counter C5:0 incremented only when the conveyor was running and Box Sensor I:1/2 produced a valid transition.

When the accumulated count reached 10, C5:0/DN energized Batch Complete Lamp O:2/1. Reset Push Button I:1/3 cleared the counter and completion indication without stopping the conveyor.

All six defined functional tests passed. The project strengthened my understanding of motor latching, process permissives, sensor-based counting, batch logic, and reset behavior, and it provides a clear foundation for a more industrial Version 2 conveyor-control architecture.